Cheatsheet AZ-104 — 06 · Blob and Files: tiers, lifecycle, versioning, soft delete, snapshots and Azure File Sync

1. Blobs and containers

Blob type	Purpose	Detail
Block blob	Files in general (images, backups, videos)	Up to ≈ 190 TiB; **the only one with access tiers**; uploaded in blocks
Append blob	Logs, append-only writes	No tiers
Page blob	VHD / disks, random access	Up to 8 TiB; no tiers

- A container is a **flat** namespace ("folders" are name prefixes) except with **hierarchical namespace** (Data Lake Gen2: real directories and ACLs). Container names: lowercase, 3-63 characters.
- Each blob has **metadata** (key-value pairs), **index tags** (up to 10, queryable with `Find blobs by tags` and usable in lifecycle), properties (Content-Type, tier, lease) and can have **snapshots** and **versions**.
- **Static website:** enabled on the account, publishes the `$web` container at `https://<account>.z<n>.web.core.windows.net` (static content only; HTTPS with a custom domain → Azure CDN / Front Door).

2. Access tiers (block blobs, GPv2 accounts)

	Hot	Cool	Cold	Archive
Use	Frequent access	Infrequent	Rare, but immediate read	Almost never; **offline**
Cost	Expensive to store, cheap to access	Cheaper to store, more expensive to access	Cheaper still	Cheapest to store; most expensive to read
Minimum retention	—	**30 days**	**90 days**	**180 days** (otherwise early deletion charge, prorated)
Latency	ms	ms	ms	**Hours**: standard rehydration up to **15 h**, high priority under 1 h (blobs < 10 GB)
Availability	99.9 %	99 %	99 %	99 %
Redundancies	All	All	All	Only **LRS, GRS, RA-GRS**
Account default tier	Yes	Yes	Yes	**No** (blob level only)

- A blob with no explicit tier shows "**Hot (inferred)**" (inherits the account's). Changing the account default tier recalculates all inferred ones (and charges operations).
- Moving to a colder tier is instant and billed as a **write**; moving to a warmer one is billed as a **read** (+ early deletion if the minimum retention was not met). To rehydrate from Archive: **Set Blob Tier** (in place) or **Copy Blob** to another blob in an online tier (avoids the penalty and keeps the original). A blob in Archive only allows reading **metadata**, not the content.
- **Premium** block blobs **have no tiers**; to move to tiers you must copy to a Standard account. Changing the tier only applies to **block blobs**.

3. Lifecycle management

```
{ "rules": [ { "enabled": true, "name": "logs", "type": "Lifecycle",
  "definition": {
    "filters": { "blobTypes": [ "blockBlob" ], "prefixMatch": [ "logs/2026" ] },
    "actions": {
      "baseBlob": {
        "tierToCool":    { "daysAfterModificationGreaterThan": 30 },
        "tierToArchive": { "daysAfterModificationGreaterThan": 90 },
        "delete":        { "daysAfterModificationGreaterThan": 365 } },
      "snapshot": { "delete": { "daysAfterCreationGreaterThan": 90 } },
      "version":  { "delete": { "daysAfterCreationGreaterThan": 90 } } } } } ] }
```

- **Filters:** `blobTypes` (blockBlob, appendBlob), `prefixMatch` (`container/prefix`), `blobIndexMatch` (tags). **Actions:** tierToCool, tierToCold, tierToArchive, delete, and `enableAutoTierToHotFromCool` (back to Hot if accessed). **Conditions:** days since modification, since creation, since **last access** (*last access time tracking* must be enabled) or since the last tier change.
- Runs **once a day**; the first run can take up to 24 h and policy changes up to 48 h. Up to **100 rules** per policy. **Does not rehydrate** from Archive. On Premium block blob accounts only delete actions (no tiering). "Minimize cost", "automatically move to archive after 180 days" → lifecycle, not a script.

4. Data protection for blobs

Feature	What it protects	Exam details
Soft delete (blobs)	Deleted or overwritten blobs, snapshots and versions	Retention **1-365 days** (default 7 when enabled from the portal); **Undelete** restores; does not protect if the account is deleted
Soft delete (containers)	Deleted containers	Retention 1-365 days; the whole container is restored; independent from the blob one

Feature	What it protects	Exam details
Versioning	Every overwrite or deletion creates an automatic **version**	Restore = **copy** (promote) the previous version; required for object replication and point-in-time restore; versions cost → combine with lifecycle
Snapshots	**Manual** read-only copy of a blob at a point in time	Snapshots must be deleted before the blob (or use "delete snapshots"); not in Archive
Point-in-time restore	Return a **range of containers** to an earlier point in time	Block blobs only; requires **versioning + change feed + soft delete**; PITR retention must be **shorter** than the soft delete one; "ransomware, restore to yesterday 10:00"
Change feed	Ordered log of changes	Prerequisite for PITR and object replication; auditing
Immutable storage (WORM)	Nobody (not even admins) can delete or modify	**Time-based retention** (unlocked for testing, **locked** irreversible) and **legal hold** (until removed); at container or **version** level; "regulatory compliance", "SEC 17a-4"
Blob inventory	Daily / weekly report of blobs and properties	"report of all blobs with their tier"
Recover the account	Deleted storage account	Up to **14 days** if the name was not reused (portal → Storage accounts → Restore)

- For "**the blob was overwritten** and we need the previous content" → versioning (or snapshots if they were being taken). For "**deleted** by mistake" → soft delete + undelete. For "**deleted container**" → container soft delete. For "**compliance, no deletions for 7 years**" → locked immutability policy.

5. Azure Files

Aspect	SSD (Premium, FileStorage)	HDD (Standard: GPv2 pay-as-you-go or FileStorage provisioned v2)
Protocols	**SMB and NFS 4.1**	SMB only
Billing	**Provisioned** (v1 / v2): you pay for what is provisioned (GiB, IOPS, MiB/s)	Provisioned v2 or **pay-as-you-go** with **Transaction optimized, Hot, Cool** tiers (more transactions → optimized; more storage → cool)
Redundancy	LRS, ZRS	LRS, ZRS, GRS, GZRS (**never RA-***)
Size	**100 TiB** (provisioned v1) or **256 TiB** (provisioned v2) per share	**100 TiB** on pay-as-you-go with any redundancy (the **large file shares** setting is only needed on very old accounts) or **256 TiB** on provisioned v2
Latency	Single-digit milliseconds	Higher
When	Databases, IOPS, NFS	Team shares, lift-and-shift, File Sync

- **Quota** per share (maximum size). Share **snapshots**: up to **200** per share, retention up to 10 years, incremental, read-only; files are restored with "Previous versions" in Windows or from the portal. **Share soft delete**: enabled by default on new accounts, 1-365 days (default 7), recovers the whole share.
- **Azure Backup for Files:** snapshot-based, in a Recovery Services vault in the **same region**; daily or hourly schedule and GFS retention (daily, weekly, monthly, yearly); restores the whole share or **individual files** (item-level) in place or to another location; enables soft delete and places a lock on the account.
- Connecting: encrypted SMB 3.x over **port 445**; "Connect" script from the portal; Linux with `cifs`. If users cannot mount from the office → the ISP blocks 445: VPN / ExpressRoute + private endpoint or Azure File Sync (cheatsheet 05 for identities).

6. Azure File Sync

Piece	What it is	Rules
Storage Sync Service	Root resource in Azure	A server registers with only **one** Storage Sync Service; the storage account must be in the **same region**
Registered server	Windows Server **2016-2025** (Full or Core) with the **agent** installed and registered	Up to **30** synced shares per server; do not clone / sysprep after installing the agent; failover cluster supported (File Server for general use)
Sync group	Sync relationship	**A single cloud endpoint** per sync group; a share can only be in **one** sync group
Cloud endpoint	The Azure file share (**SMB** only)	Changes made **directly in Azure** are detected by a job every **24 h** (server changes are uploaded within minutes)
Server endpoint	Folder on a registered server	**One per server** in each sync group; paths cannot overlap; not on the system volume for tiering
Cloud tiering	Keeps only hot files locally; cold ones remain as **pointers** (grey icon with an X, offline attribute) and are recalled on open	Policies: **volume free space** (% free to keep) and **date policy** (not accessed in N days); files > 64 KiB only; ideal for "reduce on-premises storage"

- Scenarios: **cache** a share across several offices (multi-site, all servers see the same), **replace** a file server while keeping local access, **migrate** to Azure Files without downtime, **DR** of a file server (install the agent on a new server and add it to the sync group). NTFS ACLs are preserved.
- Backup: Azure Backup of the share in the cloud, or local backup on a server **without cloud tiering** (with tiering only pointers would be copied). Simultaneous edit conflicts → the losing file is saved renamed with "Conflict". Role to register servers and create cloud endpoints: **Azure File Sync Administrator** (or Owner / Contributor).

7. Commands that show up

Task	PowerShell (Az.Storage)	Azure CLI
Container and blob	`New-AzStorageContainer -Name c1 -Context $ctx` · `Set-AzStorageBlobContent -File .\a.pdf -Container c1 -Blob a.pdf -StandardBlobTier Cool -Context $ctx` · `Get-AzStorageBlob -Container c1 -Context $ctx`	`az storage container create -n c1 --account-name st1 --auth-mode login` · `az storage blob upload -f a.pdf -c c1 -n a.pdf --tier Cool --account-name st1 --auth-mode login` · `az storage blob list -c c1`
Change tier	`$b = Get-AzStorageBlob -Container c1 -Blob a.pdf -Context $ctx; $b.BlobClient.SetAccessTier("Archive")` · rehydrate: `SetAccessTier("Hot", $null, "High")`	`az storage blob set-tier -c c1 -n a.pdf --tier Archive` · `--tier Hot --rehydrate-priority High`
Soft delete and versioning	`Enable-AzStorageBlobDeleteRetentionPolicy -ResourceGroupName RG1 -StorageAccountName st1 -RetentionDays 14` · `Enable-AzStorageContainerDeleteRetentionPolicy ... -RetentionDays 14` · `Update-AzStorageBlobServiceProperty ... -IsVersioningEnabled $true`	`az storage account blob-service-properties update --account-name st1 --enable-delete-retention true --delete-retention-days 14 --enable-container-delete-retention true --container-delete-retention-days 14 --enable-versioning true`
Restore	`Get-AzStorageBlob -Container c1 -IncludeDeleted -Context $ctx` → `$b.BlobBaseClient.Undelete()` · snapshot: `$b.BlobClient.CreateSnapshot()`	`az storage blob undelete -c c1 -n a.pdf` · `az storage blob snapshot -c c1 -n a.pdf` · `az storage container restore -n c1 --deleted-version <v>`
Lifecycle	`Set-AzStorageAccountManagementPolicy -ResourceGroupName RG1 -StorageAccountName st1 -Policy (New-AzStorageAccountManagementPolicyRule ...)` or `-Policy` with a policy object	`az storage account management-policy create --account-name st1 -g RG1 --policy @policy.json`
File share	`New-AzRmStorageShare -ResourceGroupName RG1 -StorageAccountName st1 -Name share1 -QuotaGiB 1024 -EnabledProtocol SMB` · snapshot: `New-AzRmStorageShare ... -Snapshot` · soft delete: `Update-AzStorageFileServiceProperty ... -EnableShareDeleteRetentionPolicy $true -ShareRetentionDays 14`	`az storage share-rm create --storage-account st1 -g RG1 -n share1 --quota 1024 --enabled-protocols SMB` · `az storage share-rm snapshot` · `az storage account file-service-properties update --account-name st1 --enable-delete-retention true --delete-retention-days 14`
Azure File Sync	`New-AzStorageSyncService` · `Register-AzStorageSyncServer` · `New-AzStorageSyncGroup` · `New-AzStorageSyncCloudEndpoint` · `New-AzStorageSyncServerEndpoint -ServerLocalPath D:\data -CloudTiering -VolumeFreeSpacePercent 20 -TierFilesOlderThanDays 30` · `Invoke-AzStorageSyncChangeDetection` (force detection on the cloud endpoint)	`az storagesync` (extension): `az storagesync create`, `az storagesync sync-group create`, `az storagesync sync-group cloud-endpoint create`, `az storagesync sync-group server-endpoint create --cloud-tiering on --volume-free-space-percent 20 --tier-files-older-than-days 30`

8. Decision tree

```
What does the wording ask for?
├─ Infrequently used data, immediate access, minimum cost ..... Cool (≥ 30 days) or Cold (≥ 90 days)
├─ Long retention, hours of wait acceptable ..... Archive (≥ 180 days; LRS / GRS / RA-GRS only)
├─ Read an archived blob ..... Rehydrate (Set tier or Copy Blob; high priority < 1 h)
├─ Automatically move by age / delete ..... Lifecycle management (rules by prefix)
├─ Recover a deleted blob ..... Soft delete → Undelete
├─ Recover the previous version of an overwritten blob ..... Versioning (or snapshot if one existed)
├─ Recover a deleted container ..... Container soft delete
├─ Return a whole container to a point in time (ransomware) ..... Point-in-time restore
├─ Nobody can delete or modify for N years ..... Immutable storage (time-based, locked) / legal hold
├─ Share with high IOPS or NFS ..... Azure Files Premium (SSD)
├─ Cheap share for documents ..... Azure Files Standard (HDD) Hot / Cool / Trans. optimized
├─ Share backup with per-file recovery ..... Azure Backup (snapshots) or manual snapshots
├─ Several offices share files with a local cache ..... Azure File Sync (multi-site) + cloud tiering
└─ Free up file server disk without losing files ..... Cloud tiering (volume free space / date policy)
```

9. Signals → answer

Wording	Answer
"accessed once a month, must be available immediately"	Cool
"accessed rarely, retrieval within hours is acceptable, lowest cost"	Archive
"blob in archive, need it in 1 hour"	Rehydrate high priority
"deleted blobs before 30 days incur charges"	Early deletion for Cool (30), Cold (90), Archive (180)
"move blobs to cool after 60 days, delete after 2 years, minimal effort"	Lifecycle management rule
"lifecycle rule based on last read"	Enable last access time tracking + daysAfterLastAccessTimeGreaterThan
"accidentally deleted a blob", "recover"	Soft delete (retention) → Undelete
"restore previous version after overwrite"	Blob versioning (copy the version)
"point-in-time restore prerequisites"	Versioning + change feed + soft delete (longer retention)
"WORM", "cannot be deleted even by administrators"	Immutability policy (locked)
"storage account deleted 5 days ago"	Recover (up to 14 days)
"premium file share", "NFS"	FileStorage account (SSD)
"file share larger than 5 TiB"	Large file shares (legacy setting: current accounts already reach 100 TiB)
"restore a single file from the share"	Share snapshot (manual or Azure Backup)

Wording	Answer
"branch offices need low latency access to the same files"	Azure File Sync, server endpoint in each office
"files appear with a grey X icon / offline"	Cloud tiering: they are in Azure, recalled on open
"files changed in the Azure portal take a day to appear on the server"	Change detection on the cloud endpoint every 24 h (or `Invoke-AzStorageSyncChangeDetection`)
"second server endpoint on the same server for the same sync group" fails	One server endpoint per server and sync group
"sync group with two Azure file shares"	Impossible: one cloud endpoint per sync group

10. Mistakes to avoid

1. Setting Archive as the account default tier, or on a ZRS account.
2. Expecting to read a blob in Archive without rehydrating, or expecting lifecycle to rehydrate it.
3. Enabling point-in-time restore without versioning, change feed and soft delete.
4. Confusing soft delete (deletions) with versioning (overwrites).
5. Assuming a locked immutability policy can be removed.
6. Choosing Azure Files Standard when NFS or high IOPS are required (that is Premium), or RA-GRS for Files.
7. Creating a sync group with two cloud endpoints or two server endpoints on the same server.
8. Running a local backup of a server with cloud tiering (it copies pointers).
9. Installing the File Sync agent and then cloning or running sysprep on the server.